Communication as Governance: Measuring Communicative Stance in University Intellectual Property Policies

Working Paper

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Abstract

University intellectual-property policies allocate rights and also communicate authority. We develop the Policy Communication Stance Index (PCSI), a human-anchored measure of supportive versus restrictive language, from 87,159 sentences in 518 source documents at 150 U.S. research universities (1944–2025). As scored by our classifier, written stance leans restrictive: 86 percent of university-years fall below the scale midpoint. Stance differs persistently across universities, but observable characteristics explain little of it. The index reflects supportive and obligation-oriented wording and does not reduce to readability, legal density, or measured rule form. Since 1980, stance has become more restrictive within universities, policy topics, and rule forms.

Keywords: institutional governance; policy communication; university technology transfer; intellectual property; text as data; organizational heterogeneity
JEL Codes: O34, O38, I23, C55, K11

1 Introduction

Universities govern faculty inventions through written policy. These intellectual-property policies say who owns an invention, when it must be disclosed, how rights are assigned, and how licensing revenue is shared. A large literature studies university patenting, licensing, and technology-transfer offices (Mowery et al., 2001; Sampat, 2006; Jensen et al., 2003). A related literature looks at academics' wider engagement with industry, which goes well beyond licensing (Perkmann et al., 2021).

Even the best-known policy provision has been hard to link to outcomes. Lach and Schankerman (2008) reported that universities giving inventors a larger royalty share earn more licensing income. Ouellette and Tutt (2020) later found errors in how those royalty shares were coded and no robust evidence of the effect. Coding a policy's terms, in other words, may capture only part of what the policy conveys. The same disclosure or assignment rule can be written as a command or as an offer of help. This paper measures that second dimension. It does not estimate how the wording affects inventors or technology-transfer outcomes.

We call this dimension *communicative stance*. Research on organizational communication treats formal texts as part of what makes an organization: a text can speak for an institution and shape how readers understand its authority (Smith, 2001; Cooren, 2004; Fairhurst and Putnam, 2004; Putnam and Cooren, 2004). One policy may present disclosure as a duty owed to an authority. Another may present the same requirement as a step in a supported process. Section 2 separates this framing from the content of the rules.

The distinction raises an economic question. Can the wording of a fixed rule change how costly, or how cooperative, a reader expects the process to be? Models and experiments on controlling incentives suggest it can, although they do not study university IP policies (Bénabou and Tirole, 2003; Falk and Kosfeld, 2006). Work on the expressive function of law makes a similar point about messages that go beyond formal sanctions (Sunstein, 1996). We use these ideas to motivate a simple mechanism, not to claim an observed effect. The closest existing measure, the survey-based Policy Communication Index of Canary et al. (2013), concerns communication practices and channels. Our object is narrower: the stance written into the policy text itself.

We build the Policy Communication Stance Index (PCSI) from IP policies at 150 U.S. research universities over 1944–2025. The corpus has 87,159 cleaned sentences from 518 source documents. These give 480 university-years with policy text and 4,277 annual observations of the policy in force. Human coders labeled sentences as restrictive, neutral, or supportive. A fine-tuned contextual classifier[1] extends those judgments to the full corpus, and we average its predicted probabilities to the university-year level. The approach follows the text-as-data principle of defining the concept before scaling it up with statistics (Grimmer and Stewart, 2013; Gentzkow et al., 2019; Adcock and Collier, 2001).

[1] A contextual classifier is a language model that reads the words around a phrase when deciding what the phrase means. Fine-tuning means taking a model that was already trained on general text and training it further on the sentences labeled for this study.

The paper asks three questions. First, how do universities speak? Mean PCSI is 0.437, and 86.0 percent of annual observations fall below the neutral midpoint of 0.50. In most policies, the classifier puts more weight on restrictive language than on supportive language. Validation checks show that the index tracks supportive versus obligation-oriented wording, not readability or legal density. It also does not reduce to the grammatical form of the rules, as measured by keywords: rule form explains under 5 percent of the sentence-level variation in stance.

Second, is stance changing? Average PCSI falls over the modern period, both within fixed groups of universities and in years with new policy text. The decline is not confined to ownership and assignment clauses. Disclosure, revenue-sharing, and administrative language also become more restrictive, and so does language within each rule form. Most of the decline happens within categories of clauses rather than through a shift in which clauses policies contain.

Third, who speaks differently? Stance behaves like a lasting trait of each university: 61.0 percent of the variation in the annual panel, and 48.7 percent in years with new policy text, lies between universities. Private R1 universities write more supportive policies than public R1 universities. Among policies written in the same year, the gap is about 0.04 after controlling for other characteristics. Land-grant status and other observable characteristics add little. Together they explain only a small share of the differences across universities.

The main contribution is to make a neglected feature of technology-transfer governance measurable: how formal rules address inventors. We define stance through human judgment, extend it to a large archive with a calibrated classifier, and test whether simple features of legal prose can explain it away. They cannot. Universities differ in how they speak even within the same rule forms, and that language changes over decades. Section 2 sets out the conceptual framework. Section 3 explains how PCSI is built and what it captures. Section 4 reports its level, its change over time, and its institutional pattern. Section 5 discusses interpretation and limitations, and Section 6 concludes.

2 Conceptual Framework: Policy Text as Governance

One view in organization studies holds that organizations exist in and through communication. Written texts matter in this view because they travel. They can speak for an institution long after their authors have moved on (Cooren, 2004; Smith, 2001; Putnam and Cooren, 2004; Fairhurst and Putnam, 2004). A university IP policy is such a text. It sets out what the university expects of inventors, it may stay in force for many years, and it is the university's formal statement of how invention rights will be handled. A policy text therefore does two things: it states rules, and its wording sets up a relationship between the institution and the reader.

Institutional rules state what people must or may do. The same rule can also signal how an institution expects to relate to the people it governs (Crawford and Ostrom, 1995; Phillips et al., 2004). The Institutional Grammar breaks an institutional statement into five parts: who is addressed, the obligation or permission, the action, the conditions, and any consequence for not

complying (Crawford and Ostrom, 1995). We study the language used to express these parts. It tells inventors whether they are addressed as subjects of control or as partners the institution will help, and it can vary while the rule itself stays the same. This leads to three propositions about the texts. They say nothing about how inventors respond.

P1. Separability. If framing differs from rule content, stance should vary a good deal among sentences with the same rule form. A stance measure should not reduce to counts of obligations, prohibitions, and sanctions.

P2. Institutional persistence. Texts become institutionalized when they come from actors with authority and are reused by others (Phillips et al., 2004). Studies of contracts show that formal language is often copied and changed bit by bit, leaving traces of earlier drafts (Kahan and Klausner, 1997; Choi et al., 2013). Stance should therefore be a lasting trait of each organization, and broad organizational categories should predict it only weakly.

P3. Field-level drift. Organizations in the same field respond to shared legal and professional pressures and adopt similar forms (DiMaggio and Powell, 1983; Edelman, 1992). If those pressures change the register in which universities write, the change should appear across many universities and many parts of their policies, not only in the clauses a particular court case touched.

A simple model of reader engagement (Appendix B) makes the possible economic channel explicit: two equally demanding rules might signal different expected burdens. It is an illustration, not a model we estimate.

Table 1 shows where each proposition is tested. P1 is a precondition for the measure to mean anything, and Section 3.5 tests it. Sections 4.3 and 4.2 test P2 and P3. Everything that follows tests features of the texts, not the engagement model or effects on outcomes.

Table 1: Framework Propositions and Where They Are Examined

Proposition	Observable implication	Evidence in this paper
P1. Separability	Stance varies within rule forms; PCSI is not a count of obligations	Keyword-measured rule form explains 4.8% of sentence-level stance; PCSI net of rule form correlates 0.94 with PCSI (Section 3.5)
P2. Persistence	Stance is university-specific, revised in small steps, weakly predicted by categories	About half of variation is between universities; median revision 0.021; observables explain at most a sixth (Section 4.3)
P3. Field-level drift	Decline within universities, clause topics, and rule forms	Within-university decline; most of the balanced-sample change occurs within categories (Section 4.2)

3 Measuring the Policy Communication Stance Index

The task is to turn a qualitative feature of governance, namely how a university communicates authority, obligation, and support, into a score that can be compared across universities and years. We first define the texts and the unit of measurement. We then define stance through human judgments, extend those judgments to the full corpus with a classifier, and combine the results into PCSI. Finally, we ask what the index captures.

3.1 Policy Texts and the Unit of Measurement

The corpus covers IP policies at 150 U.S. research universities between 1944 and 2025: 87,159 cleaned sentences from 518 source documents, each identified by university, year, and file. These documents govern the core of the university–inventor relationship: ownership, disclosure, assignment, licensing authority, commercialization procedures, and revenue sharing.

A university can have more than one relevant document in the same year. We then pool the sentences from all of them before computing the score. This gives 480 university-years with policy text. Between those years, we carry the most recent score forward until a newer document appears.[2] Doing this through 2025 gives 4,277 annual university-year observations of the policy in force. These are annual records of policies in force, not separately collected documents. Each keeps the year of the document its score comes from, so every analysis can separate the annual panel from the 480 years with new text. Table 2 summarizes the sample, and Appendix A describes how the panel was built.

Table 2: Corpus and Analysis Sample, 1944–2025

Quantity	Value
Universities	150
Analysis period	1944–2025
Source policy documents	518
Cleaned sentences	87,159
University-years with source policy text	480
Annual policy-in-force university-years	4,277

Notes: When several source documents are available for the same university and year, their sentences are pooled before the university-year score is constructed. In the annual panel, that score is carried forward until a newer policy source becomes available.

3.2 Defining Communicative Stance through Human Judgment

We define communicative stance as the supportive, neutral, or restrictive way a policy presents rights, obligations, authority, and assistance. The concept is about how governance is expressed. It

[2]Carrying a score forward assumes that the last observed policy stays in effect until a new version is found. It does not mean a fresh document was collected every year.

is not the generosity of the underlying rule, general positive or negative sentiment, or how faculty actually react.

Human coders turned this definition into labels.[3] Supportive sentences emphasize help, guidance, encouragement, collaboration, shared interests, or facilitation. Restrictive sentences emphasize mandatory assignment, categorical duties, unilateral authority, prohibitions, sanctions, or compliance. Neutral sentences mainly define, describe, or set out procedure, with no clear supportive or restrictive lean. Coders labeled the main message of the whole sentence, and disagreements were settled by human adjudication. The coder-level records from that stage were not kept, so we cannot report historical agreement between coders.

Rule content and stance need not line up. A disclosure requirement can be framed as a demand or as a step in a guided process. A technically dense sentence can be supportive, and a short one can assert firm control. Table 3 gives examples from the corpus. The last two rows are disclosure provisions written with opposite stance. Appendix C gives the full coding rubric.

Table 3: Illustrative Sentence-Level Communicative Stance

Class	Illustrative sentence	$\hat{p}_{\text{class}}$
Supportive	"Inventors are encouraged to consult with the OTT concerning their options regarding the release of intellectual property."	0.973
Neutral	"Special procedures relevant to these processes are posted on the website of the Office of the Vice President for Research and Innovation."	0.980
Restrictive	"As a condition of affiliation with the University, members of the University community are bound by all University policies, including this one."	0.982
Two disclosure provisions, different framing		
Supportive	"The University intends that the obligation for disclosure shall not unduly interfere with prompt publication of research results."	0.975
Restrictive	"... Each Invention must be disclosed promptly upon its creation, conception or discovery."	0.978

Notes: $\hat{p}_{\text{class}}$ is the calibrated model probability assigned to the displayed class. Source institutions and years are recorded in the replication package. The last two rows both concern invention disclosure but do not impose identical legal terms; they illustrate how provisions serving the same policy function can differ in stance. The examples illustrate the annotation construct; they are not an independent accuracy test.

3.3 Scaling the Human Judgments to the Full Corpus

Human coding defines the target, but no team can hand-label more than 87,000 sentences. We therefore use supervised machine learning to scale up the human judgments. The classifier is a fine-tuned `bert-base-uncased` model (Devlin et al., 2019). A contextual model helps here because the meaning of words such as *shall*, *may*, *support*, or *assign* depends on the words around them.

The labeled sample has 2,732 sentences: 2,458 used for training and 274 held out to test the

[3]Annotation means labeling examples by hand. Adjudication means reviewing the cases where coders disagree and settling on a final label.

model on sentences it did not see during training. On the held-out set, the original training run reports 90.1 percent accuracy and a macro-F1 score of 0.894.[4] For every sentence the model gives a probability for each of the three classes. We adjust these probabilities with temperature scaling before averaging them (Guo et al., 2017).[5] Appendix C reports the model's settings.

3.4 From Sentence Probabilities to PCSI

Let $(\hat{p}_{Ri}, \hat{p}_{Ni}, \hat{p}_{Si})$ be the adjusted probabilities that sentence i is restrictive, neutral, or supportive. We place restrictive language at 0, neutral at 0.5, and supportive at 1. The sentence score is then

$$s_i = 0.5\,\hat{p}_{Ni} + \hat{p}_{Si}. \tag{1}$$

A sentence the model is sure is supportive scores close to 1. One it is sure is restrictive scores close to 0, and one it is sure is neutral scores close to 0.5. For university j in year t, PCSI is the simple average of the sentence scores in the pooled policy text:

$$\text{PCSI}_{jt} = \frac{1}{N_{jt}} \sum_{i=1}^{N_{jt}} s_i = 0.5 + 0.5\,(\bar{\hat{p}}_S - \bar{\hat{p}}_R)_{jt}\,. \tag{2}$$

The index runs from 0 to 1. A value below 0.5 means that, on average, the policy carries more restrictive than supportive probability. A value of 0.5 does not require every sentence to be neutral; supportive and restrictive sentences can also cancel out.

3.5 What the Measure Captures

Before interpreting PCSI, we ask three questions. Does it depend on how sentence scores are combined? Which familiar features of policy language does it reflect? And does it simply restate the form of the rules a policy contains?

Aggregation. We rebuild the policy score three other ways: as the median sentence score, as a 5-percent trimmed mean, and as an average weighted by sentence length.[6] The trimmed and length-weighted versions track the baseline closely (correlations of $r = 0.999$ and 0.930).[7] The

[4] Accuracy is the share of held-out sentences whose predicted label matches the human label. For each class, precision is the share of the model's predictions of that class that are correct, and recall is the share of that class's sentences the model finds. The F1 score combines the two (it is their harmonic mean). Macro-F1 averages the three class-level F1 scores with equal weight, so the most common class cannot dominate.

[5] Temperature scaling rescales a model's probabilities with one fitted number so that they better match how often the model is right. It never changes which class has the highest probability.

[6] A 5-percent trimmed mean drops the highest and lowest 5 percent of sentence scores before averaging, which limits the influence of extreme values. The median is the middle score.

[7] The Pearson correlation r measures how closely two measures move together in a straight line, from -1 to 1. The Spearman rank correlation ρ measures the same thing for the rankings, so it asks whether two measures put universities in the same order.

median keeps most of the ranking (Spearman $\rho = 0.878$) but changes the distances between policies more (Appendix D).

Linguistic content. We build three simple diagnostics from the same text. *Tone* compares supportive and collaborative language with obligation, control, and sanction language. *Clarity* captures sentence structure and procedural guidance. *Legal Load* captures legal and IP-technical vocabulary (Appendix F). These diagnostics are not alternative definitions of stance and are not part of PCSI. If PCSI measures what we intend, it should relate most strongly to Tone and should not simply reproduce readability or legal density.

Table 4 confirms this. On its own, Tone explains 21.6 percent of the variation in PCSI.[8] Adding Clarity leaves this unchanged, and adding Legal Load raises R^2 to 0.236. Because the gain from each variable depends on the order in which variables are added, the table also reports Shapley shares (Grömping, 2006).[9] Tone accounts for 89 percent of the explained variation, Legal Load for 9 percent, and Clarity for 2 percent. In the 480 university-years with new text, Tone's share is 78 percent. Tone and Legal Load are nearly unrelated ($r = 0.029$). A policy can therefore be technically dense and still offer help, and a plain policy can assert firm control. Together the three diagnostics explain about a quarter of PCSI. The rest may reflect patterns that word lists miss, such as discretion, relational framing, or the order in which duties and help appear, as well as measurement error.

Table 4: Interpreting PCSI with Transparent Linguistic Features

	(1)	(2)	(3)
Tone Index	0.064***	0.063***	0.062***
	(0.012)	(0.012)	(0.012)
Clarity Index		0.001	0.006
		(0.008)	(0.008)
Legal Load Index			0.013**
			(0.006)
R^2	0.216	0.216	0.236
Shapley share of R^2: Tone / Clarity / Legal Load			0.89 / 0.02 / 0.09
Same, years with source text			0.78 / 0.14 / 0.09
Observations	4,277	4,277	4,277

Notes: Dependent variable is policy-in-force PCSI; 150 universities. Standard errors clustered by university are in parentheses. Shapley shares average each index's marginal contribution to R^2 over all orders of entry and sum to one. Each diagnostic combines standardized components, so one unit of a composite should not be read as one standard deviation of that composite. *** $p < 0.01$; ** $p < 0.05$; * $p < 0.10$.

[8] The share of variation explained, R^2, runs from 0 to 1. It is the fraction of the differences in PCSI across observations that a regression accounts for.

[9] A Shapley share divides a regression's explained variation among its explanatory variables. With three diagnostics there are six possible orders of entry; the Shapley share averages each diagnostic's added R^2 over all six. It describes fit, not causal importance.

Rule form versus framing. A sharper concern is that PCSI just counts rules, so that a policy full of obligations scores as restrictive however it states them. Following the Institutional Grammar of Crawford and Ostrom (1995), we use keyword rules to give each sentence a deontic operator[10]: obligation (*shall*, *must*, *is required*), prohibition (*shall not*, *may not*), permission (*may*, *is entitled*), or none. We also flag sanction language, the grammar's "or else" part. Obligations are the most restrictive form on average: their mean sentence score is 0.329, against 0.467 for sentences with no operator. But variation within forms dwarfs the differences between them. The eight combinations of form and sanction explain only 4.8 percent of the sentence-level variation in stance (Appendix Table H3). Prohibitions are no more restrictive than permissions on average, which shows how little the operator alone reveals about framing. If we subtract each sentence's form average before combining scores, the result correlates 0.94 with PCSI and gives the same headline results (Appendix I). PCSI is therefore not a count of musts and mays. The keyword indicators, however, record only operators and sanction words, not the full legal content of a sentence. So the exercise does not show that framing is fully separate from legal content.

These checks set the scope of the measure. PCSI describes formal policy text. It does not measure the generosity of legal rights, faculty perceptions, actual technology-transfer practice, or commercialization performance.

4 The Empirical Structure of Communicative Stance

We now describe the level of the index, its change over time, and its pattern across universities. The analyses are descriptive. None treats institutional characteristics or policy language as causes.

4.1 PCSI Lies Below the Neutral Midpoint

Figure 1 shows PCSI across the 4,277 annual observations, and Table 5 gives summary statistics. Mean PCSI is 0.437, the median is 0.427, and the standard deviation is 0.066.[11] The middle half of observations lies between 0.394 and 0.471, and values range from 0.264 to 0.730.[12]

The key result is where the distribution sits relative to the midpoint: 86.0 percent of annual observations fall below 0.50. Since PCSI equals $0.5 + 0.5(\bar{\hat{p}}_S - \bar{\hat{p}}_R)$, a value below 0.50 means the policy carries more restrictive than supportive probability on average. It does not mean that every sentence is restrictive or that the rules treat inventors badly. The index also varies meaningfully. A one-standard-deviation difference in PCSI equals a 0.132 difference in the average gap between supportive and restrictive probability. The upper tail reaches 0.730, so far more supportive policies are possible within the same genre.

[10] A deontic operator is the word that makes a statement an obligation, a permission, or a prohibition, such as *must*, *may*, or *must not*. Our keyword rules capture this one element, not all five parts of the Institutional Grammar.

[11] The standard deviation measures how widely values are spread around the mean. Roughly two-thirds of observations lie within one standard deviation of the mean.

[12] The middle half is the interquartile range, from the 25th to the 75th percentile. The 25th percentile is the value below which a quarter of observations fall.

The pattern does not come from carrying scores forward. Among the 480 university-years with new text, mean PCSI is 0.429 and 89.6 percent fall below 0.50. The midpoint is a reference point on the scale, not a normative benchmark. The lean also depends partly on how the classifier treats procedural language: a check against a research assistant's labels on a random sample of sentences finds a weaker lean (Appendix C).

Table 5: Distribution of PCSI

	Policy-in-force panel	Years with source text
Mean	0.437	0.429
Median	0.427	0.423
Standard deviation	0.066	0.063
25th percentile	0.394	0.391
75th percentile	0.471	0.464
Minimum	0.264	0.264
Maximum	0.730	0.730
Share below 0.50	86.0%	89.6%
Observations	4,277	480

Notes: The policy-in-force panel carries each university's most recent score forward until a newer policy source appears; the second column uses only university-years for which policy text is directly available. Analysis window 1944–2025; 150 universities.

4.2 PCSI Drifts toward Restriction over the Modern Period

Figure 2 plots mean PCSI each year from 1978, the first year with at least five universities in the data. The shaded band is a 95 percent confidence interval for each year's mean.[13] Average stance peaks around 1980 at roughly 0.49 and falls over the following decades to about 0.421 in 2025.

The raw series mixes changes in language with changes in which universities the archive covers. The decline persists when we hold the set of universities fixed. Among the 93 universities already in the data by 2000, mean PCSI falls from 0.447 to 0.422 by 2025. With university fixed effects, the trend is -0.00112 per year.[14] Its standard error, clustered by university, is 0.00042, and $p = 0.008$.[15,16] Groups of universities fixed in 1990, 1995, and 2005 show the same pattern (Appendix Table G1). Using only the 467 university-years with new text from 1980 onward, and so dropping carried-forward years, gives a trend of -0.00117 (SE $= 0.00039$, $p = 0.003$). Starting

[13] A 95 percent confidence interval is a range built so that, across repeated samples, it would contain the true value 95 percent of the time. Narrow bands mean precise estimates.

[14] University fixed effects compare each university only with itself over time. They remove stable differences between universities, so the trend reflects change within universities.

[15] A standard error measures the uncertainty of an estimate. Clustering by university allows observations from the same university to be related, which usually makes standard errors larger and more honest. It changes the uncertainty, not the estimate itself.

[16] The p-value is the probability of seeing a result at least this large if the true effect were zero. Values below 0.05 are conventionally called statistically significant.

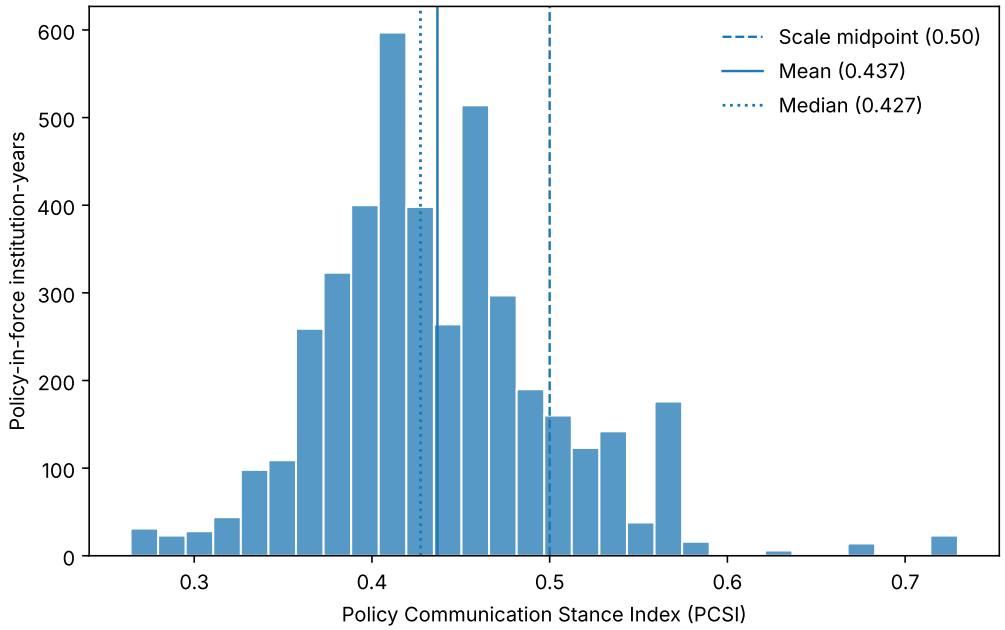


Figure 1: Distribution of PCSI in the 1944–2025 Policy-in-Force Panel ($N = 4{,}277$). The dashed line marks the scale midpoint of 0.50; the solid and dotted lines mark the mean and median.

in 1990 gives a similar estimate. The estimate from 2000 onward is negative but less precise (Appendix Table G2).

Is the decline a change in rules or in stance? One explanation is a change in the rules themselves. In *Stanford v. Roche* (2011), the Supreme Court distinguished a promise to assign patent rights from a present assignment of those rights.[17] Ownership and assignment clauses are also among the most restrictive in our corpus. If drafting changes were concentrated in those clauses, the fall in the index might mostly reflect their content. We therefore compare language within policy topics and within rule forms, without assuming that keywords fully capture legal meaning.

First, we sort sentences into clause topics with simple keyword rules: ownership and assignment, disclosure, revenue sharing, licensing and commercialization, sponsored research, copyright and software, administration and disputes, and definitions (Appendix H). For each topic we estimate the within-university trend in mean sentence score, using years with new text from 1980. Figure 3 shows that the decline is broad. The trend in ownership and assignment language is close to zero (-0.0003 per year, $p = 0.64$). Disclosure (-0.0022, $p = 0.04$), revenue sharing (-0.0017, $p = 0.01$), and administrative language (-0.0013, $p = 0.02$) all become more restrictive, and every other topic has a negative estimate.

Second, we split the decline into two parts with a shift-share decomposition.[18] One part

[17] *Board of Trustees of the Leland Stanford Junior University v. Roche Molecular Systems, Inc.*, 563 U.S. 776 (2011), https://www.law.cornell.edu/supremecourt/text/09-1159. The decision draws a distinction between assignment wordings; by itself it does not show how many universities later revised their policies.

[18] A shift-share decomposition splits a change in an average into two parts. The "within" part is the change that

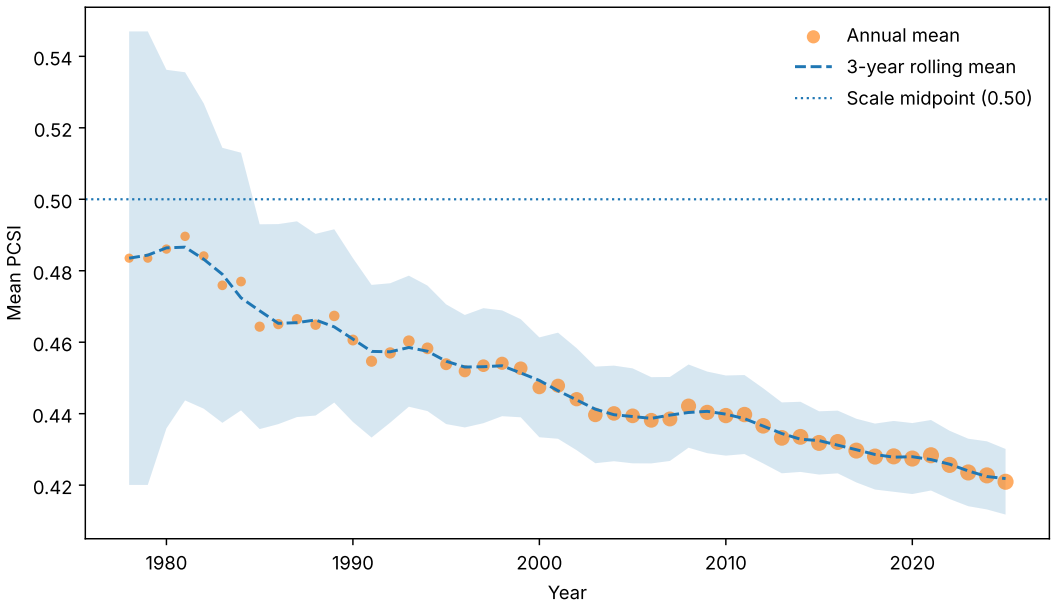


Figure 2: Annual Mean PCSI, 1978–2025. Points show each year's mean in the policy-in-force panel; point size reflects the number of universities. The shaded band is the 95% confidence interval for the annual mean, the dashed line is the three-year rolling mean, and the dotted horizontal line marks the scale midpoint of 0.50. Years before 1978 are omitted because fewer than five universities are represented.

comes from changes in stance within categories of clauses. The other comes from changes in how much text falls in each category. We use the 63 universities with new text in both 1980–1999 and 2010–2025, and each counts equally whatever its number of sentences. For each university we split its change in mean sentence score into the two parts, and then we average across universities. For topics, each sentence gets one main topic by a fixed keyword priority, and sentences with no match go into an "other" category. (The topic trends above instead allow a sentence to match several topics.) For rule form, each sentence falls into one of the eight form-and-sanction combinations. Appendix Table H1 reports the results and explains how missing categories are handled.

Mean PCSI for these universities fell by 0.02882, from 0.45545 to 0.42664. By topic, changes within categories contribute −0.02223, or 77.1 percent of the decline, and changes in category shares contribute −0.00659, or 22.9 percent. By rule form, changes within forms contribute −0.02708 (94.0 percent) and changes in the mix of forms contribute −0.00174 (6.0 percent). Both splits add up exactly to the observed change, university by university and in total.

The two splits are different ways of describing the same change. They are not effects of different causes. The split depends on the keyword priority, the balanced sample, the handling of categories missing in one period, and equal weighting of universities. Above all, a change within a keyword-defined category may still reflect a real change in the rules. The decomposition supports one limited statement: the decline is not mainly explained by changes in the shares of

would occur if each category's share stayed fixed but its average changed. The "composition" part is the change that comes from shifts in the categories' shares.

the categories we measure.

The topic results show that the modern decline is not confined to ownership and assignment clauses, and it also appears within rule forms. Because topics and rule forms are measured with keywords, these results cannot rule out substantive legal changes within categories, and the exact within-category share depends on how the categories are defined.

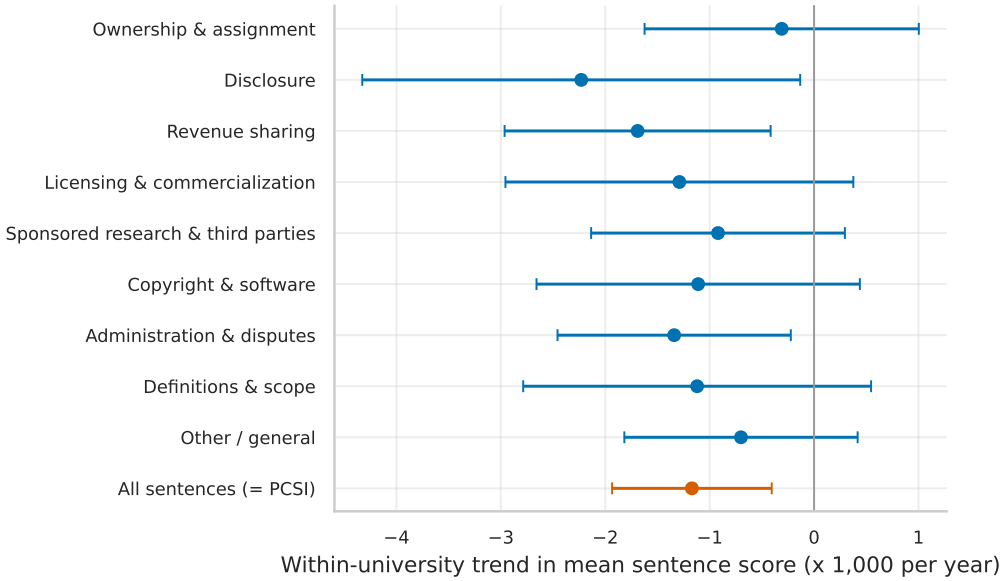


Figure 3: Within-University Trends in Sentence Stance by Clause Topic, 1980–2025. Each point is the yearly trend in a topic's mean sentence score within universities, estimated from years with new policy text from 1980 onward and universities observed in at least two such years; trends are multiplied by 1,000. Bars are 95% confidence intervals with standard errors clustered by university. Topics are assigned by keyword rules and can overlap (Appendix H).

4.3 Stance Is a Persistent Institutional Trait, Weakly Explained by Observables

Where the variation lies. We split the total variation in PCSI into a between-university part and a within-university part.[19] In the annual panel, 61.0 percent of the variation is between universities and 39.0 percent is within universities over time. Carrying scores forward mechanically raises persistence. Using only the 480 years with new text lowers the between-university share to 48.7 percent (Appendix Table D2). Carrying forward affects the size of the share but does not create the pattern. Revisions are usually small: across 330 successive policy revisions within universities, the median absolute change in PCSI is 0.021. The pattern resembles a stable institutional writing style. This is an analogy to lasting differences found in other organizations, not evidence about the people who drafted these policies (compare Bertrand and Schoar, 2003).

[19]The between-university share is the fraction of total variation in PCSI explained by differences in each university's average score. The within-university share is the rest: variation around each university's own average over time.

Who speaks differently. Next we ask whether observable characteristics explain these differences.[20] Table 6 reports university-level means. Each university contributes the average of its annual scores, so universities with longer archives do not count more. Several groups are small (there are eight private R2 universities) and the means are skewed, so we use rank-based tests.[21] Private R1 universities average 0.456, against 0.427 for public R1 universities (Kruskal–Wallis $p = 0.004$ across the four types). Land-grant universities average 0.420, against 0.442 for other universities (Wilcoxon rank-sum $p = 0.010$). Both differences survive a Holm adjustment across the four comparisons; medical-school status and urbanicity do not (Appendix Table E1).[22] Both differences keep their direction under the alternative ways of combining scores, and when university means use only years with new text (Appendix E).

Table 7 looks at the characteristics together in regressions.[23] Columns (1)–(3) use one average per university, with robust standard errors.[24] Compared with public R1 universities, the private-R1 coefficient is 0.029 (SE 0.009) with institutional type alone, 0.019 after adding land-grant, medical-school, and urbanicity controls, and 0.020 (SE 0.014) after adding region fixed effects. The drop is not caused by a change in sample: Column (1) stays at 0.029 when estimated on the 144 universities with complete data.

University averages, however, mix institutional differences with the decades in which each university's policies were written. With only 144 observations, they are also imprecise. Column (4) therefore uses the university-years with new text and adds fixed effects for the year each document was written. Each policy is thus compared only with policies written in the same year, and standard errors are clustered by university. Here the private-R1 difference is 0.040 (SE 0.012) with all controls and region fixed effects, while the land-grant coefficient is -0.005 and imprecise. Weighting university averages by the number of documents or sentences, or removing each year's

[20] R1 and R2 are research-activity categories in the Carnegie Classification of Institutions of Higher Education (methodology at https://carnegieclassifications.acenet.edu/carnegie-classification/classification-methodology/). The data contain a single yes/no `Carnegie R1` flag, and we label unflagged universities R2. This R2 group is therefore simply "not R1" within our sample, not a confirmed match to an official R2 definition. The Carnegie edition behind the flag is not recorded. The 2021 Basic Classification assigns R1 and R2 with a research-activity index. The 2025 research designations use fixed thresholds (at least $50 million in research spending and 70 research doctorates for R1; at least $5 million and 20 research doctorates for R2). The two editions are not interchangeable, so our institutional comparisons describe the supplied groups rather than official Carnegie categories. Control separates public from private universities. Land-grant universities are those designated under the Morrill Acts of 1862 and 1890. Medical-school status shows whether the university runs a medical school. Urbanicity groups campuses as rural, semi-urban, or city; without the original coding key it cannot be matched to the National Center for Education Statistics locale codes. Region groups states into six U.S. regions. All of these are fixed attributes recorded in the replication data.

[21] Rank-based tests compare groups using the ranks of the values rather than the values themselves, so they do not require normally distributed data and are not driven by a few extreme values. The Kruskal–Wallis test compares three or more groups; the Wilcoxon rank-sum test compares two.

[22] Running several tests raises the chance that at least one looks significant by luck. The Holm procedure corrects for this by ordering the p-values and comparing them with progressively less strict cut-offs. We apply it to the four institutional comparisons, not to every specification in the paper.

[23] Each regression coefficient is the average difference in PCSI associated with one characteristic, holding the others in the table fixed. For a yes/no characteristic such as land-grant status, it is the difference between universities with and without that characteristic.

[24] Robust (HC3) standard errors remain valid when the spread of the unexplained part of PCSI differs across universities. HC3 is a version that works well in small samples.

Table 6: University-Level PCSI by Institutional Characteristics

Group	*N*	Mean	SD	Range
Carnegie classification and control				
Private R1	38	0.456	0.046	[0.337, 0.534]
Public R1	86	0.427	0.051	[0.328, 0.585]
Private R2	8	0.406	0.053	[0.342, 0.459]
Public R2	18	0.441	0.058	[0.376, 0.580]
Mission and program				
Land-grant	47	0.420	0.047	[0.338, 0.585]
Non-land-grant	103	0.442	0.053	[0.328, 0.580]
Medical school	118	0.438	0.053	[0.328, 0.585]
No medical school	32	0.422	0.047	[0.342, 0.519]
Urbanicity				
Rural	15	0.425	0.041	[0.372, 0.519]
Semi-urban	48	0.422	0.047	[0.338, 0.539]
City	82	0.443	0.057	[0.328, 0.585]

Notes: Unit of analysis is the university; PCSI is averaged across each university's policy-in-force years before group statistics are computed. Five universities lack urbanicity data. Rank-based tests are reported in Appendix E.

average before computing them, does not change the pattern (Appendix Table E2).

Among these comparisons, the private-R1 difference is the clearest pattern. The land-grant difference shrinks once overlapping characteristics are taken into account. Even so, observable characteristics explain at most about a sixth of the variation in average PCSI across universities. Broad institutional categories organize some of the variation in how universities write, but they do not determine it.

5 Discussion

The striking result is not that mandatory language is common. It is that the way universities express authority differs widely across institutions and becomes more restrictive over time, even within similar kinds of provisions. Our contribution is measurement and description. These differences are not evidence that wording itself changes commercialization outcomes.

The engagement model in Appendix B gives one reason two wordings of the same rule might be experienced differently. Our data cannot say whether that happens. We observe policy wording, not whether readers understood it, disclosed inventions, sought help, patented, or licensed. A test would need a drafting experiment, or a policy revision linked to measured faculty responses.

Persistence with weak observable correlates is what P2 predicts if stance reflects each institution's drafting history. Studies of corporate contracts and sovereign bonds find that legal language is often carried over and adapted from earlier documents (Kahan and Klausner, 1997; Choi et al., 2013). This is one plausible reason for persistence in university policies, although those studies

Table 7: Institutional Correlates of PCSI

	University means			Source-text years
	(1)	(2)	(3)	(4)
Private R1	0.029***	0.019	0.021	0.040***
	(0.009)	(0.012)	(0.014)	(0.012)
Private R2	−0.021	−0.031	−0.035	−0.006
	(0.021)	(0.023)	(0.023)	(0.018)
Public R2	0.014	0.007	0.011	0.005
	(0.015)	(0.020)	(0.018)	(0.011)
Land-grant		−0.014	−0.011	−0.005
		(0.012)	(0.013)	(0.012)
Medical school		0.014	0.010	0.006
		(0.011)	(0.011)	(0.009)
Urbanicity controls	No	Yes	Yes	Yes
Region fixed effects	No	No	Yes	Yes
Source-year fixed effects	No	No	No	Yes
R^2	0.074	0.111	0.161	0.291
Observations	150	145	144	452
Universities	150	145	144	144

Notes: Dependent variable is PCSI. Columns (1)–(3) use one observation per university (mean policy-in-force PCSI) with HC3 robust standard errors; Columns (2)–(3) omit universities with missing covariates. Column (4) uses university-years with directly available policy text, compares policies written in the same year through source-year fixed effects, and clusters standard errors by university. Public R1, rural, and Midwest are the omitted categories. *** $p < 0.01$; ** $p < 0.05$; * $p < 0.10$.

do not examine universities. Shared legal and professional pressures could help explain the shift toward more restrictive wording (DiMaggio and Powell, 1983; Edelman, 1992). Drift within several keyword-defined topics and rule forms fits a change in drafting register. The decompositions attribute most of the modern decline to changes within categories under our definitions. They do not show that the legal content stayed the same, and they do not identify causes. The institutional results fit the wider technology-transfer literature, which stresses differences in professional routines, relationships between faculty and technology-transfer offices, and local commercialization practice (Owen-Smith and Powell, 2001; Siegel et al., 2003; Bercovitz and Feldman, 2008; Perkmann et al., 2021). PCSI adds a measurable textual dimension to those differences.

The study has several limits. The annual panel assumes the most recent policy stays in force until a newer one is found. Restricting the analysis to years with new text reduces reliance on that assumption and gives the same central results, but it cannot prove the archive is complete. The held-out test set is small and does not show accuracy separately by institution, period, or clause type. The classifier reads sentences one at a time, so it misses layout, headings, order, and links between passages. The keyword rules for topics and rule forms are transparent but coarse: they pick up operators and subject matter, not the full legal content of each provision. Our regressions use predicted rather than observed labels. The main results hold when PCSI is built from the single most likely label or only from sentences the model classifies with at least 90 percent probability, and the model's confidence is stable across decades (Appendix I). Formal corrections for using predicted labels in regressions exist, but each requires its own assumptions and data (Egami et al., 2023; Battaglia et al., 2024). A check against a research assistant's labels shows that the classifier gives a direction to procedural text more often than a human reader does (Appendix C). Most fundamentally, written stance can differ from legal rights and from practice. A higher PCSI is not a welfare ranking, a measure of legal generosity, or a case for weakening legitimate disclosure and ownership rules.

The replication package contains the code that scores sentences, combines the scores, builds the annual panel, and produces every table, along with the scored sentences and the model's metadata. The headline statistics have been recomputed from the sentence-level data with code written separately from the original analysis. These checks confirm the released scores and the calculations built on them; they do not retrain the classifier. Because the original annotation records were not kept, the classifier's accuracy figures come from the original training run.

6 Conclusion

This paper makes the communicative stance of university IP policy measurable. PCSI anchors supportive and restrictive language in human judgment, scales it up with a calibrated classifier, and yields a panel covering 150 research universities over eight decades. The picture is one of stable differences between institutions alongside slow change across the whole field. As the classifier scores them, most policies lean toward restrictive language. That lean is not explained

by the mix of obligations and permissions: rules of the same form can be written in very different registers. Since the 1980s, the register has become less supportive within universities and within several kinds of clauses. The measure describes wording, not the generosity of legal rights or the effectiveness of technology-transfer offices. It offers a consistent way to compare how universities communicate their rules, without treating differences in wording as differences in outcomes.

Data and Code Availability

The scored sentences, derived indices, analysis code, and tables are available in the GitHub replication repository.

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A Data Construction and Sample

The analysis uses policy documents from 150 U.S. research universities over 1944–2025: 87,159 cleaned sentences from 518 source documents, each identified by university, year, and file. When a university has more than one relevant document in a year, we pool their sentences before computing its score. This gives 480 university-years with policy text. The annual panel treats the most recent score as in force until a newer document appears for the same university. Carrying scores forward through 2025 gives 4,277 annual observations, each tagged with the year of the document behind its score. For comparisons across universities, we average each university's annual scores so that every university counts once. The archive also holds a single one-sentence record from 1925. We keep it for completeness but exclude it from the analysis window.

B A Simple Model of Reader Engagement

There is an economic reason to separate a rule from how it is communicated, even when the legal requirement does not change. Consider an inventor deciding whether to contact the technology-transfer office. Let r summarize the rule, s the stance the inventor *perceives* in its wording, and c_i the inventor's cost of seeking guidance. Suppose the net benefit of engaging rather than staying passive is

$$U_i(1) - U_i(0) = v(r) - c_i + \theta s + \varepsilon_i. \tag{3}$$

Here $v(r)$ captures the legal and practical consequences of the rule, and θs captures any change in the expected burden or help that the wording signals. The term ε_i is a random taste shock. If it follows a logistic distribution with scale $\sigma > 0$, the probability that the inventor engages is

$$\Pr(a_i = 1 \mid r, s, c_i) = \Lambda\left(\frac{v(r) - c_i + \theta s}{\sigma}\right), \qquad \Lambda(x) = \frac{1}{1 + e^{-x}}. \tag{4}$$

The function Λ turns the net benefit into a probability between 0 and 1.[25] Holding the rule fixed, more perceived support raises engagement if $\theta > 0$, has no effect if $\theta = 0$, and could lower it if the language seems untrustworthy ($\theta < 0$). We test none of these predictions. The testable implication here is narrower: *the same rule form can carry different stance*. That is the separability question the corpus can answer. A reader experiment could hold the legal provision fixed, randomly vary supportive versus commanding wording, and measure understanding, perceived burden, and willingness to ask for guidance. Such a design would test the mechanism directly.

[25] This is the standard logit model of a yes/no choice. It is used here only to illustrate the mechanism; we do not estimate it. We do not assume the sign of θ. The variable s is the stance the reader perceives; PCSI, which is built from the text, is at most a proxy for it, and we observe no behavior.

C Annotation, Model, and Replication

Coding rubric. Coders label each sentence supportive, neutral, or restrictive. Supportive language emphasizes help, guidance, encouragement, collaboration, shared interests, or facilitation. Restrictive language emphasizes mandatory assignment, unilateral authority, prohibitions, sanctions, forfeiture, or categorical compliance. Neutral language mainly defines, describes, or sets out procedure, with no clear lean. Coders label the main message of the whole sentence. Several trained coders applied the rubric, and a human adjudicated disagreements. The preserved files contain 2,732 adjudicated sentences. The coder-level records from before adjudication were not kept, so agreement between the original coders cannot be reported.

Check against a second reader. After the analysis was complete, a research assistant labeled a random sample of 400 corpus sentences with the same rubric. An AI coder (Claude, a large language model given the rubric and the sentence text) labeled the same sentences independently, without seeing the classifier's output. The research assistant and the AI coder agreed on 79.5 percent of sentences (Cohen's $\kappa = 0.56$).[26] The classifier's label matched the research assistant's for 64.5 percent of sentences ($\kappa = 0.37$). Most disagreements involve sentences the research assistant judged neutral, often procedural text, which the classifier labeled restrictive or supportive. On this sample the research assistant's labels average 0.51 on the PCSI scale, against 0.44 for the classifier. The restrictive lean reported in Section 4.1 should therefore be read as a property of the classifier's scoring; a reader who treats procedural language as neutral would see a weaker lean. The coded sample is available from the authors on request.

Classifier and calibration. Table C1 reports the classifier's settings. We apply temperature scaling before averaging probabilities. The replication package reproduces the model's predictions from the saved classifier, the adjusted sentence scores, the aggregation, the panel, and the analyses; the original training run cannot be rebuilt. Rescoring a random sample of 2,000 sentences with the saved model reproduces the stored probabilities to within 4×10^{-6} and gives every sentence the same label. Every sentence score can also be rebuilt from the stored probabilities, and rebuilding the university-year indices from those scores reproduces them exactly.

D Measurement Robustness

Table D1 compares the baseline with the median, the 5-percent trimmed mean, and the length-weighted mean. Midpoint agreement is the share of observations that the baseline and the alternative place on the same side of 0.50. Trimming changes almost nothing. Length weighting stays close to the baseline. The median produces the largest change in levels but keeps a Spearman

[26] Cohen's κ measures agreement between two raters after removing the agreement expected by chance. A value of 0 means chance-level agreement and 1 means perfect agreement; values between 0.41 and 0.60 are conventionally read as moderate agreement.

Table C1: Preserved Classifier Configuration

Parameter	Value
Base model	`bert-base-uncased`
Transformer layers	12
Attention heads	12
Hidden size	768
Intermediate size	3,072
Maximum input length	256 tokens
Output classes	3
Training sentences	2,458
Validation sentences	274
Total labeled sentences	2,732
Temperature parameter	$T = 0.8462$
Validation accuracy	0.901
Validation macro-F1	0.894

Notes: Configuration and metrics are read from the preserved model artifacts in the replication package.

correlation of 0.878. Table D2 repeats the split into between- and within-university variation for each version, in the annual panel and in the 480 years with new text. Carrying scores forward raises measured persistence in every version, but substantial differences across universities remain without it.

Table D1: Sensitivity to Alternative Sentence Aggregation Rules

Aggregation	Mean	Pearson r	Spearman ρ	Midpoint agreement
Baseline (unweighted mean)	0.437	1.000	1.000	1.000
Median PCSI	0.471	0.623	0.878	0.937
5% trimmed mean	0.431	0.999	1.000	1.000
Length-weighted mean	0.433	0.930	0.906	0.933

Notes: $N = 4{,}277$ policy-in-force university-years. Correlations are with baseline PCSI. Midpoint agreement is the share of observations that the baseline and the alternative place on the same side of 0.50.

E Institutional Tests

Table E1 reports the four rank-based comparisons with Holm-adjusted p-values. The simple differences are stable across ways of combining scores. The private-R1 minus public-R1 difference is 0.029 with baseline PCSI, 0.018 with the median, and 0.032 with both the trimmed and the length-weighted mean. The land-grant minus non-land-grant difference ranges from -0.020 to -0.024. When university averages use only years with new text, the two differences are 0.033 and -0.019.

Table E2 reports the private-R1 and land-grant coefficients across samples, weights, and

Table D2: Between- and Within-University Variation

Specification	Between share	Within share
Policy-in-force panel		
Baseline	0.610	0.390
Median	0.434	0.566
Trimmed mean	0.616	0.384
Length-weighted mean	0.589	0.411
Years with source text		
Baseline	0.487	0.513
Median	0.300	0.700
Trimmed mean	0.494	0.506
Length-weighted mean	0.463	0.537

Notes: Shares partition the total sum of squared deviations of PCSI into between- and within-university components.

Table E1: Institutional Comparisons with Multiple-Testing Adjustment

Characteristic	Test	Raw p	Holm p
Carnegie classification and control	Kruskal–Wallis	0.004	0.014
Land-grant status	Wilcoxon rank-sum	0.010	0.029
Urbanicity	Kruskal–Wallis	0.055	0.111
Medical-school status	Wilcoxon rank-sum	0.172	0.172

Notes: University-level mean PCSI ($N = 150$; 145 for urbanicity). The Holm adjustment is applied to this family of four tests.

adjustments for the era of each policy. The weighted rows use weighted least squares, which gives more influence to universities with more documents or sentences.[27] The timing of the archives differs modestly by type: the average document year is 2002.3 for private R1 universities and 2005.1 for public R1 universities. Because PCSI falls over time, comparing university averages without adjusting for era slightly favors universities with older archives. That is why Column (4) of Table 7 compares policies written in the same year. The private-R1 difference is larger, not smaller, in that comparison.

Table E2: Private-R1 and Land-Grant Coefficients Across Samples and Estimators

	Type only	Full controls		
	Private R1	Private R1	Land-grant	N
University means	0.029***	0.021	−0.011	144
	(0.009)	(0.014)	(0.013)	
University means, common sample	0.029***	0.021	−0.011	144
	(0.009)	(0.014)	(0.013)	
WLS, weight = source policy years	0.027**	0.020	−0.021	144
	(0.010)	(0.015)	(0.015)	
WLS, weight = sentences	0.021	0.023	0.005	144
	(0.014)	(0.023)	(0.028)	
Era-adjusted university means	0.029***	0.022	−0.009	144
	(0.009)	(0.014)	(0.013)	
Source-text years, year FE	0.040***	0.040***	−0.005	452
	(0.010)	(0.012)	(0.012)	
Source-text years from 1980, year FE	0.039***	0.038***	−0.005	439
	(0.010)	(0.012)	(0.012)	

Notes: Dependent variable is PCSI; public R1 is the omitted type. "Full controls" adds land-grant, medical-school, and urbanicity indicators and region fixed effects. University-mean rows use HC3 standard errors; the published row uses 150 universities for the type-only column and 144 with full controls. Era-adjusted means subtract each calendar year's cross-sectional mean from the policy-in-force panel before averaging within university. Source-text rows use university-years with directly available policy text, include source-year fixed effects, and cluster standard errors by university. *** $p < 0.01$; ** $p < 0.05$; * $p < 0.10$.

F Transparent Linguistic Diagnostics

The diagnostics help interpret the classifier-based measure. They are not independent tests of how readers perceive policies, and they are not alternative definitions of stance. Let $z(\cdot)$ be a

[27] Weighted least squares is a regression in which some observations count more than others. Here the weights are the number of source documents or sentences behind each university's average.

standardized policy-level component.[28] The indices are

$$\text{Tone} = \tfrac{1}{6}\big[z(\text{supportive}) + z(\text{second person}) + z(\text{inclusive we}) - z(\text{restrictive}) - z(\text{sanction}) - z(\text{obligation modal share})\big], \tag{5}$$

$$\text{Clarity} = \tfrac{1}{3}\big[-z(\text{mean sentence length}) - z(\text{long-sentence share}) + z(\text{procedural share})\big], \tag{6}$$

$$\text{Legal Load} = \tfrac{1}{2}\big[z(\text{legalese}) + z(\text{IP technical})\big]. \tag{7}$$

The supportive, restrictive, sanction, procedural, legalese, and IP-technical components come from word lists in the replication code. The second-person and inclusive-"we" components capture direct and collective forms of address. Components are adjusted for policy length where relevant and standardized before they are combined. The three diagnostics are distinct: Tone and Clarity correlate at 0.143, Tone and Legal Load at 0.029, and Clarity and Legal Load at -0.296. Table F1 shows that adding the log of word count or sentence count leaves the Tone relationship almost unchanged while weakening the Legal Load coefficient. In the 480 years with new text, the three-index regression gives coefficients of 0.038 for Tone, 0.013 for Clarity, and 0.012 for Legal Load, with $R^2 = 0.151$.

Table F1: Linguistic Decomposition with Policy-Length Controls

	Baseline	+ Log words	+ Log sentences
Tone Index	0.062***	0.059***	0.059***
	(0.012)	(0.012)	(0.012)
Clarity Index	0.006	0.006	0.008
	(0.008)	(0.008)	(0.009)
Legal Load Index	0.013**	0.010	0.010
	(0.006)	(0.007)	(0.007)
Policy-length control	No	Log words	Log sentences
R^2	0.236	0.244	0.241
Observations	4,277	4,277	4,277

Notes: University-clustered standard errors in parentheses. *** $p < 0.01$; ** $p < 0.05$; * $p < 0.10$.

G Temporal Robustness

The annual series can change because language changes or because the set of universities changes. Table G1 holds the set fixed, using the universities already in the data by each start year. Every group has a lower mean in 2025 than at its start, and every within-university trend is negative. Table G2 drops carried-forward years and uses only years with new text. It reports both a pooled

[28] Standardizing a variable means subtracting its mean and dividing by its standard deviation. The result, a z-score, has mean 0 and standard deviation 1, which lets components measured in different units be added together.

trend, which compares all observations together, and a fixed-effects trend, which compares each university only with itself. The 2025 cross-section is not driven by unusually old policies. Among the 70 universities whose 2025 policy is at most five years old, 91 percent fall below the midpoint, against 92 percent of all 150.

Table G1: PCSI Trends Within Fixed University Cohorts

Cohort start	Universities	Start mean	2025 mean	FE trend per year	p
1990	34	0.461	0.421	−0.00109	0.019
1995	53	0.454	0.422	−0.00131	0.002
2000	93	0.447	0.422	−0.00112	0.008
2005	114	0.439	0.420	−0.00103	0.011

Notes: Each cohort contains the universities already represented in the policy-in-force panel by the start year. Trends come from regressions of PCSI on year with university fixed effects; standard errors are clustered by university.

Table G2: Temporal Trends Using Only Years with Source Policy Text

Start year	N	Pooled trend	FE trend	SE	p
1980	467	−0.00115	−0.00117	(0.00039)	0.003
1990	443	−0.00121	−0.00128	(0.00053)	0.016
2000	367	−0.00109	−0.00150	(0.00093)	0.105

Notes: University-years with directly available policy text from the start year onward. Standard errors are clustered by university.

H Clause Topics and Rule Forms

Clause topics. Sentences are sorted into topics by keyword rules that ignore capitalization: ownership and assignment (*assign*, *ownership*, *title*, *vest*, *belong*); disclosure (*disclos*, *invention report*); revenue sharing (*royalt*, *revenue*, *net income/proceeds*, *distribut*, *equity*); licensing and commercialization (*licens*, *commerciali*, *technology transfer*, *start-up*, *patent application*); sponsored research and third parties (*sponsor*, *grant*, *federal*, *government*, *contract*, *third part*); copyright and software (*copyright*, *software*, *courseware*, *scholarly work*); administration and disputes (*appeal*, *dispute*, *committee*, *review*, *waive*, *grievance*, *amend*); and definitions and scope (*means*, *defined*, *applies to*, *purpose*, *scope*). A sentence can belong to several topics, and the topic trends keep those overlaps. For the decomposition only, each sentence gets one main topic, the first match in the order listed, and sentences with no match are classified as other/general. Keep this difference in mind when comparing the two sets of results. Among sentences from 1980 onward, 59 percent match at least one topic. Ownership and assignment sentences are the most restrictive (mean score 0.335), and definition and revenue-sharing sentences the least (0.448 and 0.453). Table H2 reports the topic trends.

Exact within-category and composition accounting. The decomposition compares sentences from 1980–1999 with those from 2010–2025 for the 63 universities with new text in both periods (8,353 early and 28,175 late sentences). For university i in period $t \in \{0, 1\}$, let s_{ikt} be the share of its sentences in category k and m_{ikt} the category's mean stance score. The university's mean is $M_{it} = \sum_k s_{ikt} m_{ikt}$. Using the average of the two periods' shares and means gives an exact identity:

$$M_{i1} - M_{i0} = \underbrace{\sum_k \frac{s_{ik1} + s_{ik0}}{2} (m_{ik1} - m_{ik0})}_{\text{within-category change}} + \underbrace{\sum_k \frac{m_{ik1} + m_{ik0}}{2} (s_{ik1} - s_{ik0})}_{\text{composition change}}. \tag{8}$$

If a category is missing for a university in one period, we set its missing mean equal to its mean in the other period. This assigns the appearance or disappearance of a category to composition rather than to within-category change. The choice is a convention, not an estimate of what would have happened. We apply the identity within each university and then average the two parts across universities with equal weights. Averaging shares and means separately across universities and then multiplying them would not give the same answer. Both decompositions add up to the observed change to within 10^{-11}. The topic version uses one main topic per sentence; the rule-form version uses the eight form-and-sanction combinations. The two versions are alternatives and should not be added together. Per-university contributions and the code that checks the identity are in the replication package.

Table H1: Exact Shift–Share Decompositions of the Modern PCSI Decline

	Exclusive primary topic	Deontic × sanction
Early period: mean score (1980–1999)	0.455454	0.455454
Late period: mean score (2010–2025)	0.426638	0.426638
Observed change	−0.028816	−0.028816
Within-category contribution	−0.022229	−0.027077
Composition contribution	−0.006587	−0.001739
Within-category share of decline	77.1%	94.0%
Composition share of decline	22.9%	6.0%
Universities with policies in both periods	63	63
Early and late sentences	8,353; 28,175	8,353; 28,175

Notes: The unit of comparison is a university appearing in both periods; universities are weighted equally. Each sentence is assigned to one primary topic by fixed keyword priority, or to one of eight deontic-by-sanction cells. Category changes are decomposed within each university using midpoint shares and midpoint conditional means, then averaged. If a category is absent in a period, its conditional mean is imputed from the observed period, assigning category entry/exit to composition. The university-level accounting identity holds to machine precision. These alternative decompositions are not additive and are sensitive to the coding rules and missing-category convention. The numbers are descriptive, not causal estimates.

Rule forms. Following Crawford and Ostrom (1995), each sentence gets one deontic operator, with this order of priority: prohibition (a negated *shall*, *must*, *may*, *will*, or *can*; *cannot*; *prohibit*; *not permitted*), then obligation (*shall*, *must*, *is required*, *requires*, *obligat-*, *agrees to*, *is expected to*), then permission (*may*, *permit*, *entitled*, *discretion*, *can*, *allow*), then none. Sanction language (*violat-*, *disciplin-*, *sanction*, *forfeit*, *terminat-*, *breach*, *penalt-*, *dismiss*, *revok-*) is flagged separately. Table H3 reports stance by form. At the policy level, the shares of obligation, prohibition, permission, and sanction sentences together explain 11 percent of the variation in PCSI across the 480 years with new text.

Table H2: Within-University Trends in Sentence Stance by Clause Topic, 1980–2025

Topic	Share	Mean score	FE trend per year	SE	p
Ownership & assignment	0.142	0.335	−0.00031	(0.00067)	0.644
Disclosure	0.057	0.357	−0.00223	(0.00107)	0.037
Revenue sharing	0.131	0.453	−0.00169	(0.00065)	0.010
Licensing & commercialization	0.133	0.432	−0.00129	(0.00085)	0.129
Sponsored research & third parties	0.131	0.341	−0.00092	(0.00062)	0.140
Copyright & software	0.092	0.412	−0.00111	(0.00079)	0.159
Administration & disputes	0.141	0.425	−0.00134	(0.00057)	0.018
Definitions & scope	0.083	0.448	−0.00112	(0.00085)	0.185
Other / general	0.409	0.427	−0.00070	(0.00057)	0.213
All sentences (= PCSI)	1.000		−0.00117	(0.00039)	0.002

Notes: Each row regresses the university-year mean sentence score within a topic on year and university fixed effects, using source-text university-years from 1980 and universities observed in at least two such years; standard errors are clustered by university. Topics are assigned by keyword rules and can overlap; share is the fraction of 1980–2025 sentences matching the topic. Within-topic trends allow overlapping keyword matches. The exact shift-share calculation instead uses mutually exclusive primary topics and is reported separately in Table H1.

Table H3: Communicative Stance by Rule Form

Rule form	Share	Mean score	Restrictive	Supportive
Obligation	0.314	0.329	0.466	0.105
Prohibition	0.072	0.430	0.470	0.316
Permission	0.138	0.431	0.406	0.246
No deontic operator	0.475	0.467	0.226	0.160
Sanction language (any form)	0.037	0.334	0.487	0.124

Notes: Deontic operators follow the Institutional Grammar of Crawford and Ostrom (1995) and are assigned by keyword rules with priority prohibition > obligation > permission (Appendix H). Sanction language (the grammar's "or else" component) can co-occur with any form. Restrictive and supportive shares use the classifier's most likely label. Rule form explains 4.8 percent of sentence-level variance in stance. The exact 63-university within-cell and composition changes are reported in Table H1.

I Prediction Uncertainty and Alternative Versions of PCSI

PCSI averages probabilities rather than single labels, so it keeps track of the classifier's uncertainty. Our regressions, however, still treat the predicted scores as data. Egami et al. (2023) propose a method that uses extra human-coded labels to correct inference, and Battaglia et al. (2024) study regressions whose variables are generated from text. Their corrections differ, and we implement neither. Instead we check directly whether the main results depend on uncertain predictions. Table I1 recomputes them three ways. The first uses hard labels, scoring each sentence 0, 0.5, or 1 by its single most likely class. The second uses only the 66 percent of sentences whose most likely class has a probability of at least 0.9. The third uses the measure net of rule form from Section 3.5. Every version puts most policies below the midpoint, attributes about half of the variation in years with new text to universities, shows a negative within-university trend, and gives a positive and significant private-R1 coefficient. The model's confidence is also stable over time. From the 1970s on, the average probability of the most likely class lies between 0.874 and 0.886 in every decade, so older text is not scored with more uncertainty.

Table I1: Headline Results Under Alternative Constructions of PCSI

Construction	r with baseline	Mean	Share < 0.50	Between share	FE trend, 1980+	Private R1
Baseline PCSI		0.429	0.896	0.487	−0.00117***	0.040***
					(0.00039)	(0.012)
Hard labels	0.980	0.425	0.856	0.487	−0.00139***	0.042***
					(0.00044)	(0.013)
High-confidence sentences	0.952	0.418	0.875	0.495	−0.00134***	0.049***
					(0.00051)	(0.014)
Framing net of rule form	0.943	0.427	0.894	0.492	−0.00135***	0.032**
					(0.00037)	(0.013)

Notes: All columns use the 480 university-years with directly available policy text. Hard labels score each sentence 0, 0.5, or 1 by its most likely class. High-confidence uses only sentences whose largest calibrated class probability is at least 0.9 (about two-thirds of sentences). Framing net of rule form subtracts the mean score of each sentence's deontic-by-sanction cell before aggregation. The trend is the coefficient on year with university fixed effects, using universities with at least two source-text years from 1980; the private-R1 coefficient is from the Column (4) specification of Table 7. Standard errors clustered by university. *** $p < 0.01$; ** $p < 0.05$; * $p < 0.10$.